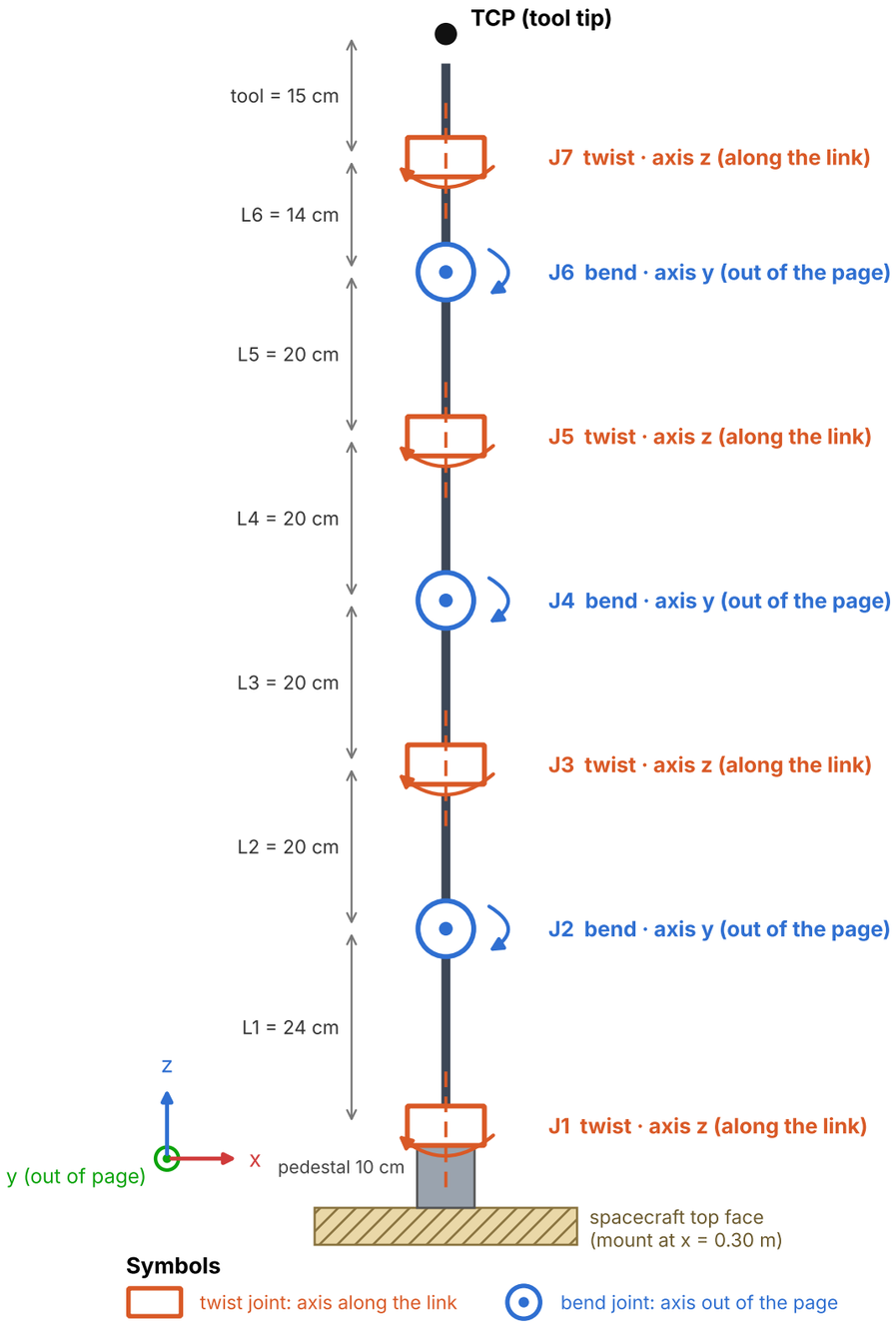
STP_Spacearm 7-DOF arm — the axis of every joint

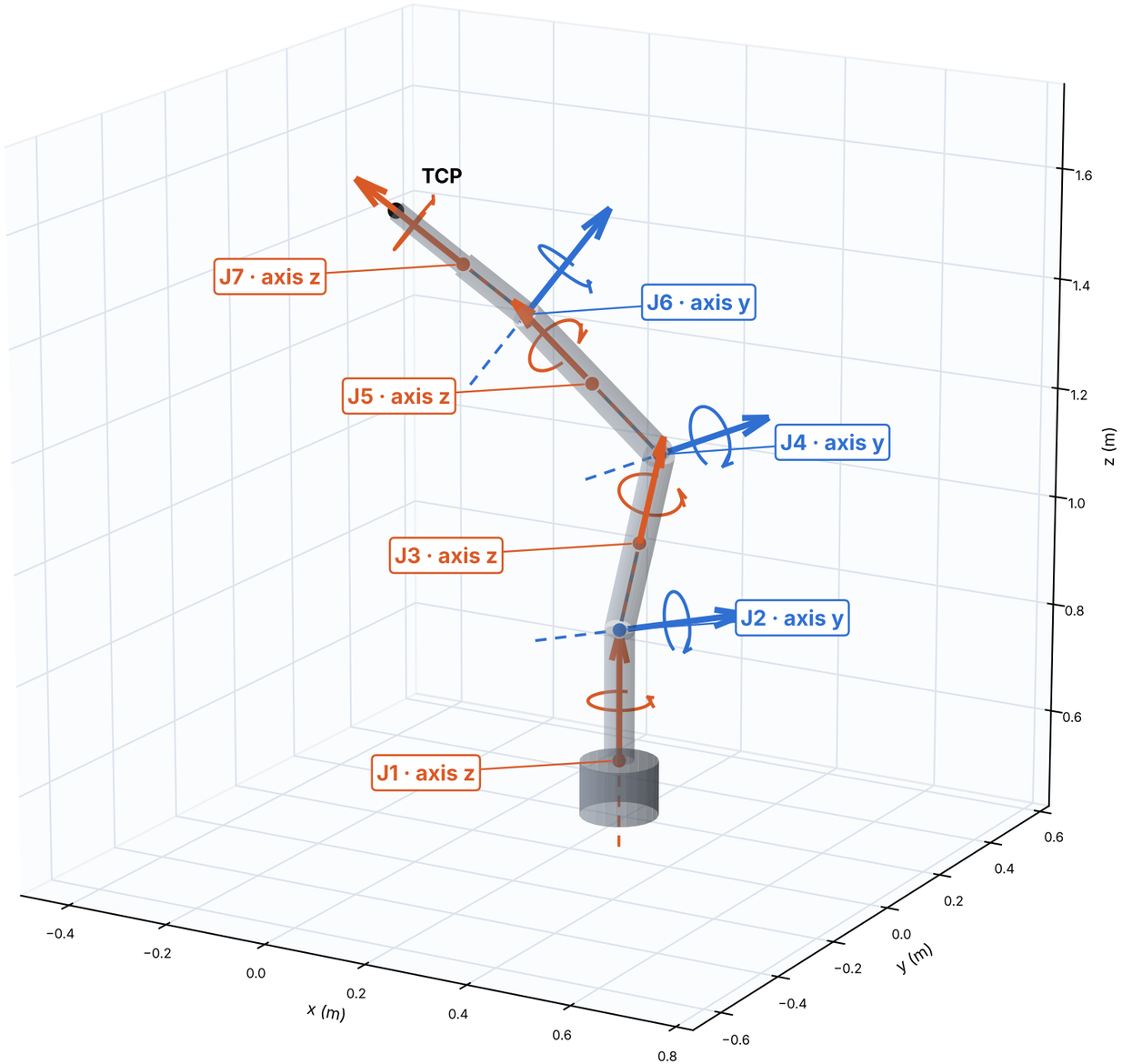
Axis order z · y · z · y · z · y · z (shoulder J1–J3 · elbow J4 · wrist J5–J7). Each axis is expressed in the frame of the link before it; at q = 0 all z axes point up and all y axes point along +y.

A · Kinematic schematic (zero configuration, side view)



B · Every joint axis in 3-D (example pose)

q = (−37, 25, 15, −88, −55, 34, 0)°



Solid arrow = positive rotation axis (right-hand rule: thumb along the arrow, fingers show +q). Dashed line = the full axis through the joint.

Limits J1: z [-170°, 170°] J2: y [-120°, 120°] J3: z [-170°, 170°] J4: y [-120°, 120°] J5: z [-170°, 170°] J6: y [-120°, 120°] J7: z [-175°, 175°] · Source: akashrayganguly/STP_Spacearm configs/default.yaml (commit a15a269); drawn from the configuration, no project code run.